



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 22: Oligopoly

Patricia Galilea A.

ICS2523 Microeconomics

Lecture 22

Last lecture, we studied Monopoly

- Regulation
- Price discrimination
 - First-degree
 - Second-degree
 - Third-degree

In today's lecture, we will start studying **Oligopoly**:

- Definitions: Best response, Nash Equilibrium, dominant strategy
- Backward induction
- Review examples of matrix games
- Example of continuum of actions → Tragedy of the Commons

Third Assignment: Essay Writing (16th June)

The **third assignment** is to write an essay (2500 words) about Transantiago (choosing between two subjects):

- Analyse the economic incentives involved in [fare evasion](#)
- Should Transantiago (Red) have more or less [subsidies](#)?

Each subject has a recommended scientific paper, which you may use as a starting point. However, you should also look for more information yourself.

I expect a critical analysis using your microeconomic knowledge and your ideas. Fare evasion is a theme more suited for Consumer Theory analysis, whereas incentives should be more related to Welfare, Consumer Theory and Producer Theory.

Third Assignments: Essay Writing (16th June)

The **second assignment** covered the first steps:

- Choose your **question + justification**
- Build a **structure** for your essay
- Write the **introduction**
- **Prompts** used and **enhanced prompts** in any AI tool, with their justifications → **We will grade them, again** 🤔

You are **allowed to use an AI tool** for:

- Generating ideas for research questions
- Improving writing quality
- Formatting citations and references
- Improving structure
- Translating parts of your writing

Homework Lecture 21

Suppose that a monopoly producer of widgets has a constant marginal cost of $c = 6$ and sells its products in two separated markets whose inverse demand functions are:

$$P_1 = 24 - Q_1 \text{ and } P_2 = 12 - \frac{Q_2}{2}$$

If the monopolist uses Third-degree price discrimination,

- a) Which are the profit-maximizing price-quantity combinations in these two markets?
- b) Find the profits
- c) Calculate the total deadweight loss (compared with the competitive output)
- d) If the monopolist is constrained by the government to charge a single price, what will happen with welfare?

Homework Lecture 21: Solution

$$P_1 = 24 - Q_1 \text{ and } P_2 = 12 - \frac{Q_2}{2}$$

If the monopolist uses a Third-degree price discrimination,

- a) **Which are the profit-maximizing price-quantity combinations in these two markets?**

In each market, $MR = MC = 6$

$$\begin{aligned} MR_1 &= \frac{\partial P_1}{\partial Q_1} \cdot Q_1 + P_1 = -1 \cdot Q_1 + 24 - Q_1 = 24 - 2Q_1 = 6 \\ &\rightarrow Q_1^* = \frac{18}{2} = 9, P_1^* = 15 \end{aligned}$$

$$\begin{aligned} MR_2 &= \frac{\partial P_2}{\partial Q_2} \cdot Q_2 + P_2 = -\frac{1}{2} \cdot Q_2 + 12 - \frac{Q_2}{2} = 12 - Q_2 = 6 \\ &\rightarrow Q_2^* = 6, \quad P_2^* = 9 \end{aligned}$$

Homework Lecture 21: Solution

b) Find the profits

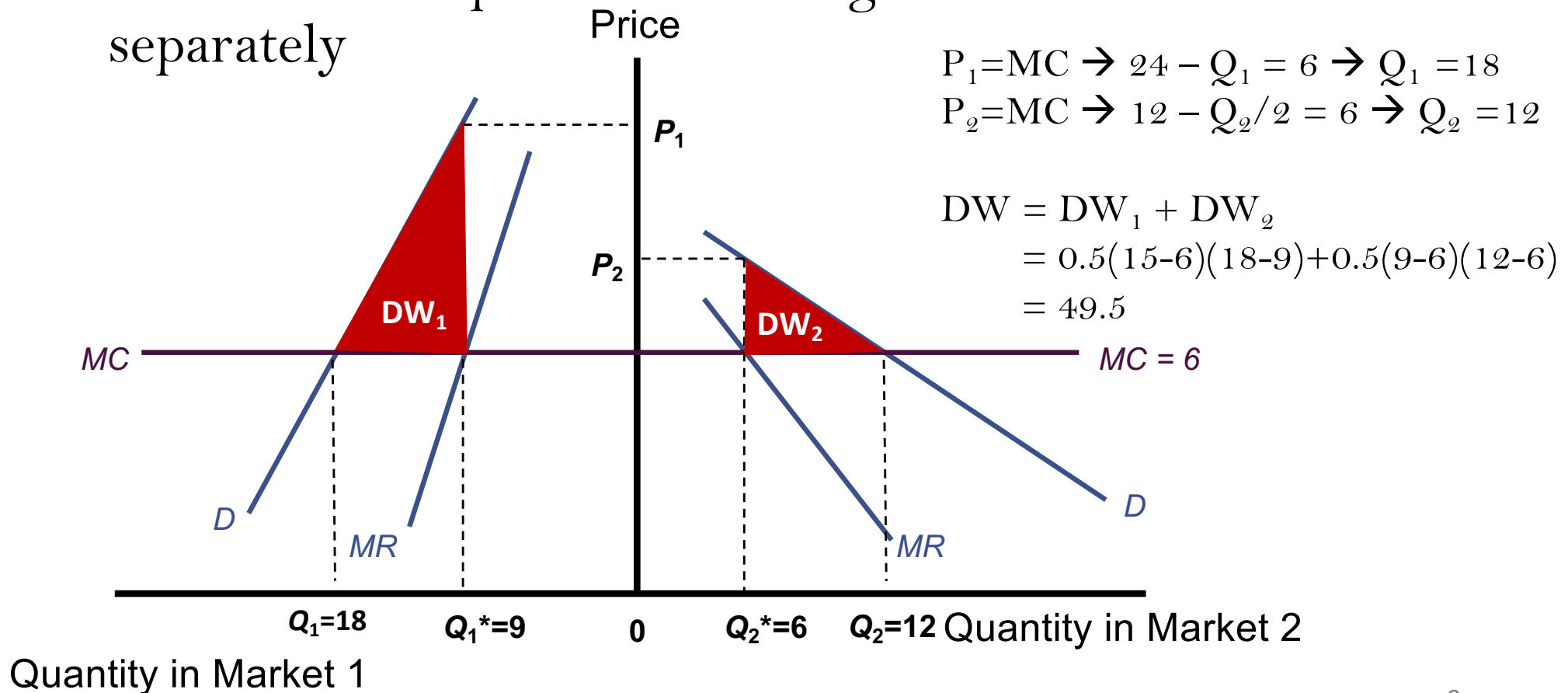
Profits are:

$$\begin{aligned}\pi &= P_1^* \cdot Q_1^* + P_2^* \cdot Q_2^* - C(Q_1^* + Q_2^*) \\ \pi &= 15 \cdot 9 + 9 \cdot 6 - 6 \cdot 15 = 99\end{aligned}$$

Homework Lecture 21: Solution

c) Calculate the total deadweight loss (comparing with the competitive output)

We should compute the deadweight losses in the two markets separately



Homework Lecture 21

- d) If the monopolist is constrained by government to charge a single price, what will happen with welfare?

In this case, constraining the monopoly to charge a single price would reduce welfare.

First, we must find the total demand curve, then find the marginal revenue curve and find $MR=MC$.

The total demand curve has a kink at $P=12$: for prices below \$12 there is demand from both markets, for prices above \$12, there is only demand from the first market:

$$D_T = D_1(P) + D_2(P) = \begin{cases} 24 - P, & 12 \leq P < 24 \\ 48 - 3P, & 0 \leq P < 12 \end{cases}$$

Homework Lecture 21

- d) If the monopolist is constrained by government to charge a single price, what will happen with welfare?

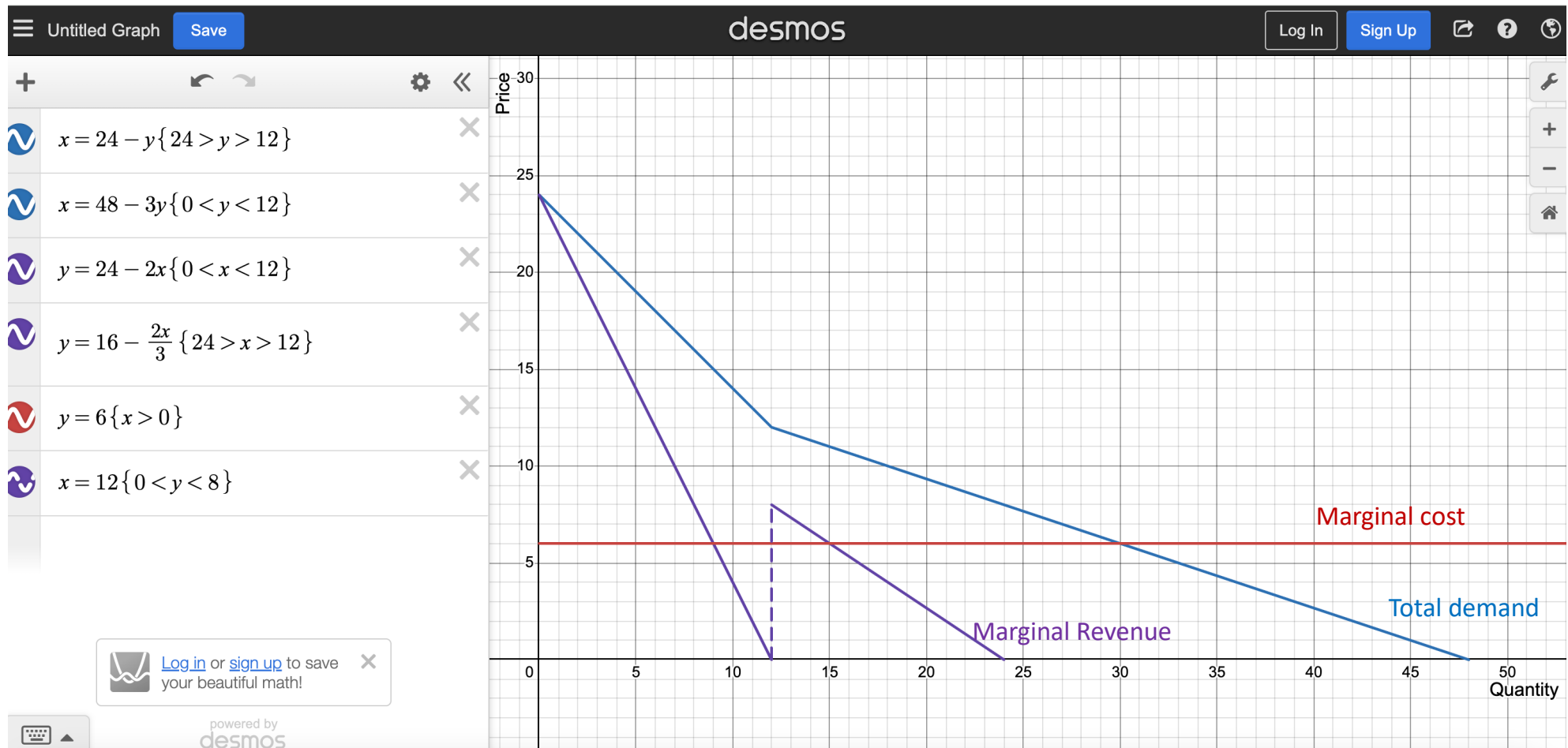
Marginal revenue uses the **inverse demand function** (demand in terms of quantity):

$$D_T^{-1}(Q) = \begin{cases} 24 - Q, 0 < Q \leq 12 \\ 16 - \frac{Q}{3}, 12 < Q \end{cases}$$

Thus,

$$MR(Q) = \begin{cases} 24 - 2Q, 0 < Q \leq 12 \\ 16 - \frac{2Q}{3}, 12 < Q \end{cases}$$

Homework Lecture 21



Homework Lecture 21

d) If the monopolist is constrained by government to charge a single price, what will happen with welfare?

The monopolist has two options:

1. $MC = 6 = 24 - 2Q \rightarrow Q = 9 \rightarrow P = 15 \rightarrow \pi = 15 \cdot 9 - 6 \cdot 9 = 81$ (serving only first market)
2. $MC = 6 = 16 - \frac{2Q}{3} \rightarrow Q = 15 \rightarrow P = 9 \rightarrow \pi = 15 \cdot 9 - 6 \cdot 15 = 45$ (serving both markets)

Thus, chooses $Q=9$, $P=15$.

Under a single-price policy the monopoly would simply cease serving market 2, since it can maximize profits by charging a price of 15 and at that price no widgets will be bought in market 2 (because the maximum willingness to pay is 12).

$$DW = DW_1 + DW_2 = 40.5 + 0.5(12-6)(12-0) = 76.5$$

Oligopolies

Market Failures

Imperfect competition

Other markets do not fit into a perfectly competitive market definition or a monopoly—they have elements of both types of markets and do not fit either model

→ Imperfect competition

One type of imperfect competition is **oligopolies**, a market structure in which few sellers offer similar or identical products —examples: cereals, detergent, OPEP, and cigarettes. Economists measure the market dominance exerted by a small number of companies with a statistic called the **concentration ratio**, which is the percentage of total output offered in the market by the four largest companies.

Another type of imperfect competition is **monopolistic competition**, a market structure where numerous companies sell similar, but not identical, products. These markets have three characteristics: many sellers, product differentiation, free entry and exit of the market —examples: computer games, restaurants, cookies.

Video monopolistic competition:

<https://www.youtube.com/watch?v=T3F1Vt3IyNc>

Imperfect competition: example

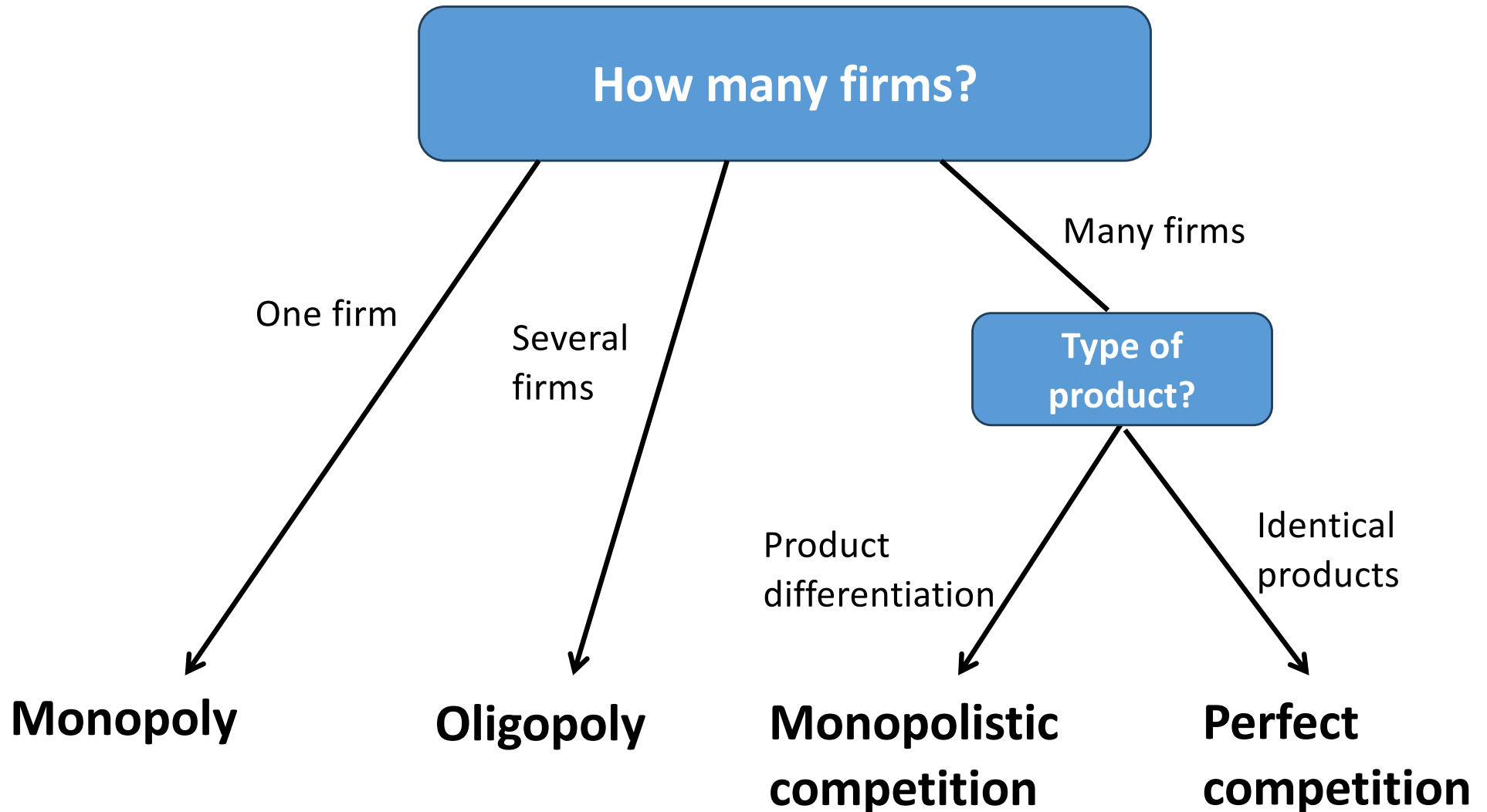
How would we describe the market of books?

On the one hand, the **book market may seem like a perfectly competitive market**: there are lots of authors, publishers, and titles, each one competing for our attention. There are a lot of options competing with each other. There are not so many barriers to enter (you have to write a book and publish it)

On the other hand, this **market seems like a monopoly**: each book is unique, so publishers have some liberty in deciding its price (they are not price takers, they are price makers). And its price is higher than the marginal cost (a hard-cover book may cost \$25, and its marginal cost is around \$5).

The book market fits better with the concept of **monopolistic competition**.

Imperfect Competition: Types of market structures



Oligopoly definition

Market Failures

7. Market Failures: Oligopoly

What's an **oligopoly**?

Market with **relatively few firms** (but more than one) that falls between the extremes of perfect competition and monopoly

Oligopolies raise the possibility of strategic interaction among firms

- If the interaction is **non-cooperative / aggressive** → Game-Theory appears (Bertrand, Cournot and Stackelberg models)
- If the interaction is **cooperative** → Cartel

Before going any further with these types of interaction, we must see some definitions about **game theory**

7. Market Failures: Oligopoly

What's an **oligopoly**?

There are **two major tasks** involved when using game theory to analyze an economic situation

- STEP 1: Distill the situation into a simple game (simple setting, retaining essential elements)
- STEP 2: Solve the given game (result is a prediction of what will happen)

Game Theory

definitions

Market Failures

8. Game Theory definitions

Game: is an abstract model of a strategic situation.

It has three essential elements:

1. **Players**
2. **Strategies and**
3. **Payoffs**

In complicated settings, it is sometimes necessary to specify additional elements (sequence of moves, information players have)

8. Game Theory definitions

1. **Players:** each decision maker (individuals, firms, nations). A player can choose from among a set of possible actions.
2. **Strategies:** each course of action is open to a player during a game. Strategies could be a simple action or a complex plan, depending on the game → We will denote S_i the set of strategies open to player i and $s_i \in S_i$ the particular strategy chosen by player i from the set of possibilities. A strategy profile will refer to a listing of particular strategies chosen by each player of a group of players
3. **Payoffs:** final returns to the players at the conclusion of a game. Payoffs are measured in levels of utility obtained by the players
→ in a two-player game, $u_1(s_1, s_2)$ denotes player 1's payoff given that he chooses s_1 and the other player chooses s_2

8. Game Theory definitions

Note: the fact that the player's 1 payoff may **depend** on 2's strategy is where the strategic interdependence shows up.

In a n-player game, payoff of a generic player $i \rightarrow u_i(s_i, s_{-i})$ which depends on player i 's own strategy s_i and the profile $s_{-i} = (s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_n)$ of the strategies of all players other than i .

8. Game Theory definitions

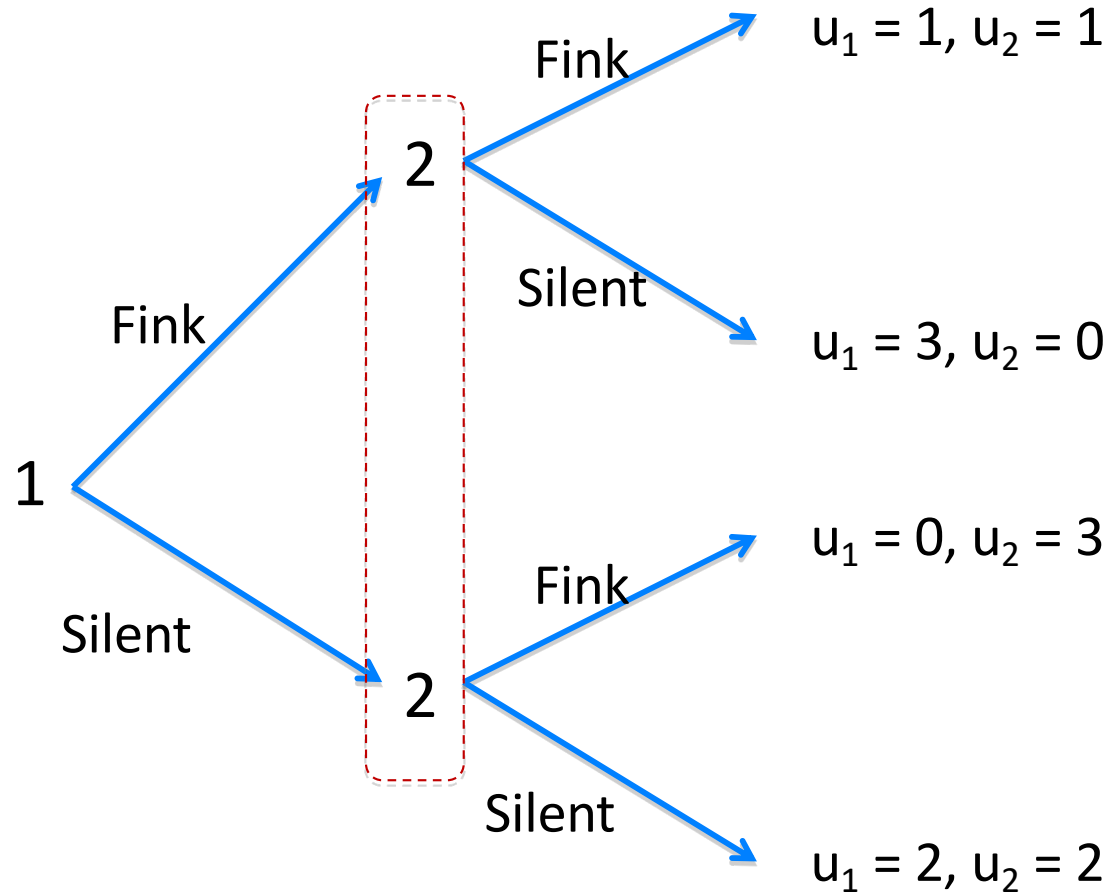
Example: Prisoners' Dilemma

Two suspects are arrested for a crime, the district attorney has little evidence in the case and it is eager to extract a confession, so she separates them and tells each:

“If you **fink** on your companion, but your companion doesn't fink you, I can promise you a reduced (one-year) sentence, whereas your companion will get four years. If you both fink on each other, you will each get a three-year sentence.”

(Each suspect knows that if neither of them finks, then the lack of evidence will result in being tried for a lesser crime → two-year sentence)

Extensive form (game tree) for Prisoners' Dilemma



Payoffs represent years of freedom during the next 4 years

Normal form (matrix) for Prisoner's Dilemma

		Suspect 2	
		Fink	Silent
Suspect 1	Fink	$u_1=1, u_2=1$	$u_1=3, u_2=0$
	Silent	$u_1=0, u_2=3$	$u_1=2, u_2=2$

Payoffs represent years of freedom during the next 4 years

Activity 1

		Suspect 2	
		Fink	Silent
Suspect 1	Fink	$u_1=1, u_2=1$	$u_1=3, u_2=0$
	Silent	$u_1=0, u_2=3$	$u_1=2, u_2=2$

Payoffs represent years of freedom during the next 4 years

What might we predict will happen?

Author:
Nicolás
Errázuriz
(2021)



8. Game Theory definitions

Best response: In a n-player game, strategy s_i is a best response to rivals' strategies s_{-i} if player i cannot obtain strictly higher payoff with any other strategy s'_i given than rivals are playing s_{-i}

$$s_i \in BR_i(s_{-i})$$

if

$$u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i}), \forall s'_i \in S_i$$

Dominant strategies: A strategy that is a best response to any strategy the other players might choose is called a dominant strategy

$$\rightarrow s_i^* \in BR_i(s_{-i}), \forall s_{-i}$$

Players do not always have dominant strategies

8. Game Theory definitions

Nash equilibrium: involves strategic choices that, once made, provide no incentives for the players to alter their behaviour. For an n -player game, a Nash equilibrium is a strategy profile $(s_1^*, s_2^*, \dots, s_n^*)$ such that for each player $i = 1, 2, 3, \dots, n$, s_i^* is the best response to the other players' equilibrium strategies $s_{-i}^* \rightarrow s_i^* \in \text{Br}_i(s_{-i}^*)$

- Nash equilibriums are stable $\rightarrow$ even if all players reveal their strategies to each other, no player would have an incentive to deviate from his equilibrium strategy.
- Nash equilibrium is guaranteed to exist for all games we will study (allowing **mixed strategies**, but we are going to study only **pure strategies**)
- But, Nash equilibrium might not be unique... making it hard to come up with a unique prediction.

8. Game Theory definitions

There is a difference between a **dominant strategy** and a **Nash equilibrium**:

- A **strategy that is part of a Nash equilibrium** needs only to be a best response to one strategy profile of other players (their equilibrium strategies)
- A **dominant strategy** must be a best response not just to the Nash equilibrium strategies of other players, but to ALL the strategies of those players

When a dominant strategy exists → it is the unique Nash Equilibrium

8. Game Theory definitions

If all players of a game have a dominant strategy $\rightarrow$ the game has a dominant strategy equilibrium (and it is a Nash equilibrium)

Let's review these definitions with the prisoners' dilemma:

		Suspect 2	
		Fink	Silent
Suspect 1	Fink	<u>$u_1=1, u_2=1$</u>	<u>$u_1=3, u_2=0$</u>
	Silent	$u_1=0, \underline{u_2=3}$	$u_1=2, u_2=2$

(Fink, Fink) is the Nash equilibrium

A quick way to find Nash equilibria is to UNDERLINE the best response payoffs in the matrix.

If player 2 finks, player 1 BR is to fink

If player 2 remains silent, player 1 BR is to fink

If player 1 finks, player 2 BR is to fink

If player 1 remains silent, player 2 BR is to fink

Activity 2: Battle of the Sexes

A wife and a husband would like to go on an evening out. They can go either to the movies or to a football game. Both prefer to spend time together rather than apart. Conditional on being together, the wife prefers to go to the movies and the husband to watch a football match.

1. Write the extensive form/game tree of this game
2. Write the normal form/matrix of this game
3. What is/are the Nash Equilibrium?
4. Is there a dominant strategy?

(The classical example is opera/ballet vs. boxing)

8. Game Theory definitions

Strictly dominated strategies elimination

	D	E	F
A	(0,4)	(7,0)	(5,3)
B	(2,0)	(2,8)	(4,7)
C	(3,3)	(3,5)	(5,6)

In this example (x,y) represent positive payoffs

- For player 1 (choosing between A, B, and C): strategy B is always dominated by C $\rightarrow$ Nash equilibrium cannot have B
 - Player 2 (choosing between D, E, and F): strategy E is dominated by F
 - If player 2 chooses D $\rightarrow$ player 1 chooses C $\rightarrow$ AD cannot be a N.E.
 - If player 1 chooses C $\rightarrow$ player 2 chooses F $\rightarrow$ CD cannot be a N.E.
 - If player 1 chooses A $\rightarrow$ player 2 chooses D $\rightarrow$ AF cannot be a N.E.
- $\rightarrow$ N.E. is CF

Example Rock, Paper, Scissors

Rock, Paper, Scissors (*Cachipún*) is a children's game in which the two players simultaneously display one of three hand symbols

	Rock	Paper	Scissors
Rock	(0,0)	(-1, <u>1</u>)	(<u>1</u> ,-1)
Paper	(<u>1</u> ,-1)	(0,0)	(-1, <u>1</u>)
Scissors	(-1, <u>1</u>)	(<u>1</u> ,-1)	(0,0)

In this example (x,y) represent positive payoffs

None of the nine boxes represents a Nash Equilibrium (in pure strategies, there is one N.E. in mixed strategies)

Any strategy pair is unstable because it offers at least one of the players an incentive to deviate.

8. Game Theory definitions

You may see a real game theory game on TV:

<https://www.youtube.com/watch?v=S0qjK3TWZE8>

- **Sequential games:** In some games, the order of moves matters. The big difference with simultaneous games is that a player that moves later in the game can observe how others have played up to that moment
 - Player can use this information to form more sophisticated strategies than simply choosing an action → he may have a contingent plan (with the action played depending on what the others have done)
- **Backward induction:** process of solving the equilibrium by working backwards (from the end of the game to the beginning)

8. Game Theory definitions

- **Repeated games:** In many real-world settings, players play the same game over and over again.
 - The simple constituent game that it is played repeatedly is called the *stage game*.
 - Repeated play of the stage game opens up the possibility of cooperation in equilibrium
 - Players can adopt *trigger strategies* (they cooperate as long as all have cooperated, but revert to playing the Nash equilibrium if anyone deviates from cooperation)
- **Finitely versus Infinitely repeated games**
 - Finitely repeated games: if the stage game has a unique Nash equilibrium, then the unique subgame-perfect equilibrium of the finitely repeated game is to play the Nash equilibrium every period (Selten's Theorem). If the stage game has multiple Nash equilibria, it may be possible to achieve some cooperation
 - Infinitely repeated games: Players may end up cooperating

8. Game Theory definitions

- **Incomplete information:** some players have information about the game that others do not
 - Examples?
 - These are complicated games, players that lack full information about the game will try to use what they do know to make **inferences** about what they do not → Bayesian games (use of bayesian rule)
 - Incomplete information games can also be sequential → player 1 moves first (*signaling games*)
- **Continuum of actions:** some settings are more realistically modeled through a continuous range of actions. We will see this further with Bertrand (players select a positive price) and Cournot (players select a positive quantity) models.



Continuum of actions: Example

The Tragedy of the Commons

Two herders decide how many sheep to graze on the village commons

Problem: commons is quite small and can rapidly succumb to overgrazing

Continuum of actions: Example

The Tragedy of the Commons

Let q_i be the number of sheep that herder $i = 1, 2$ grazes on the commons

Per-sheep value of grazing on the commons (in terms of wool and sheep-milk) is: $v(q_1, q_2) = 120 - (q_1 + q_2)$

First step: write down the payoff for each player as a function of all player's actions

$$u_1(q_1, q_2) = q_1 v(q_1, q_2) = q_1(120 - (q_1 + q_2))$$

$$u_2(q_1, q_2) = q_2 v(q_1, q_2) = q_2(120 - (q_1 + q_2))$$

Continuum of actions: Example

The Tragedy of the Commons

Second step: compute the first-order condition associated with each player's payoff maximum $\rightarrow$ find the **best response of each player** as a function of all other player's actions

N players $\rightarrow$ n equations for the n unknown equilibrium actions.

To find the Nash equilibrium we solve Herder 1's value-maximization problem

$$\max_{\{q_1\}} q_1(120 - (q_1 + q_2))$$

Continuum of actions: Example

The Tragedy of the Commons

$$\max_{\{q_1\}} q_1(120 - (q_1 + q_2))$$

First order condition:

$$120 - 2q_1 - q_2 = 0 \rightarrow q_1 = 60 - q_2/2 = BR_1(q_2) \text{ (herder 1's best response)}$$

Similarly for herder 2: $\rightarrow q_2 = 60 - q_1/2 = BR_2(q_1)$ (herder 2's best response)

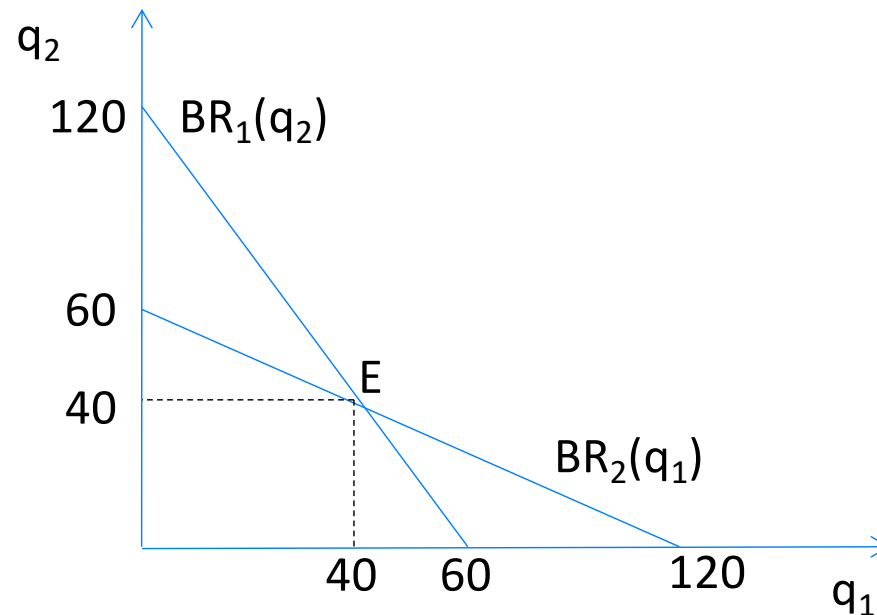
The Nash equilibrium is given by the pair (q_1^*, q_2^*) that solve the best response equations $\rightarrow (40, 40)$ and each herder earns a payoff of 1,600

PROBLEM: Nash eq is NOT the best use of the commons (less efficient)

Continuum of actions: Example

The Tragedy of the Commons

PROBLEM: Nash eq is NOT the best use of the commons (less efficient)



Joint Payoff maximization $\rightarrow (30, 30)$ gives 1,800 each

Continuum of actions: Example

The Tragedy of the Commons

Joint payoff maximization

$$\max_{\{q_1\}} (q_1 + q_2)(120 - (q_1 + q_2))$$

First order condition:

$$(120 - q_1 - q_2) - q_1 - q_2 = 0 \rightarrow q_1 = 60 - q_2 = BR_1(q_2) \text{ (herder 1's best response)}$$

Similarly for herder 2: $\rightarrow q_2 = 60 - q_1 = BR_2(q_1)$ (herder 2's best response)

This joint Payoff maximization is solved by $q_1 = q_2 = 30$ or, more generally, by any q_1 and q_2 that sum to 60. Each herder earns a payoff of 1,800. Nevertheless, this does not happen naturally.

Backward Induction

Market Failures

13. Backward Induction

Backward induction is the process of solving for equilibrium by working backward from the end of the game to the beginning.

Backward induction works as follows:

1. **Identify all the subgames** at the bottom of the extensive form
2. **Find the Nash equilibria** on these subgames
3. **Replace** the (potentially complicated) subgames with the **actions and payoffs resulting from the Nash equilibrium** play on these subgames
4. Then, **move up to the next level of subgames** and repeat the procedure

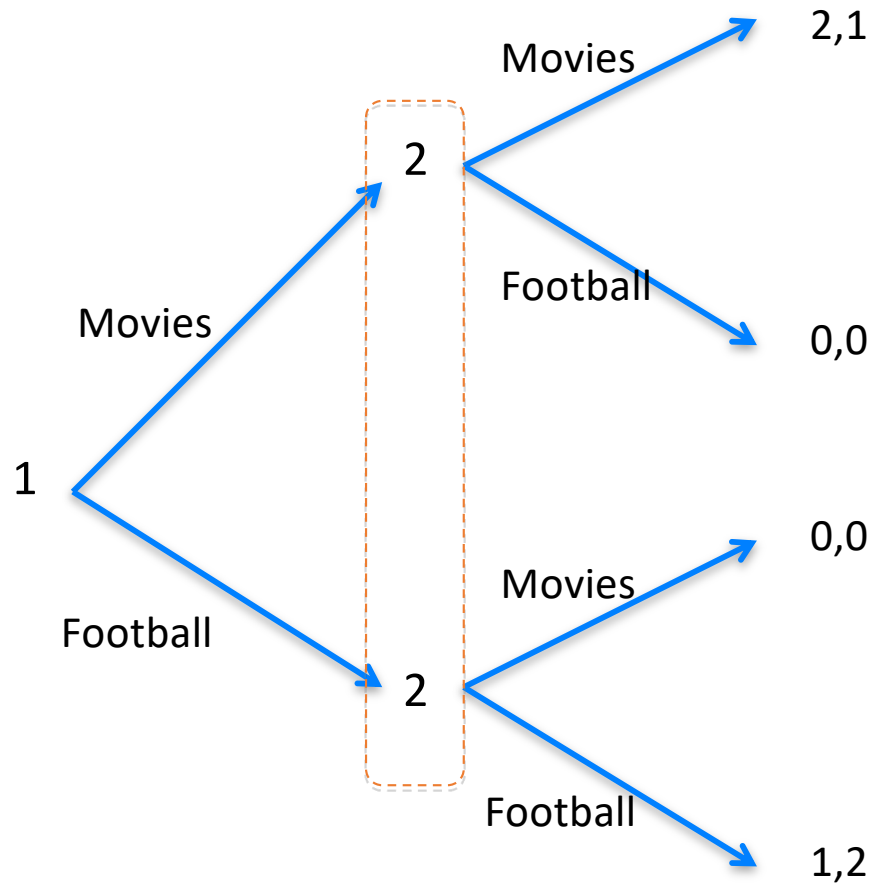
What is a subgame?

A **subgame** is a part of the extensive form beginning with a decision node and including everything that branches out to the right of it.

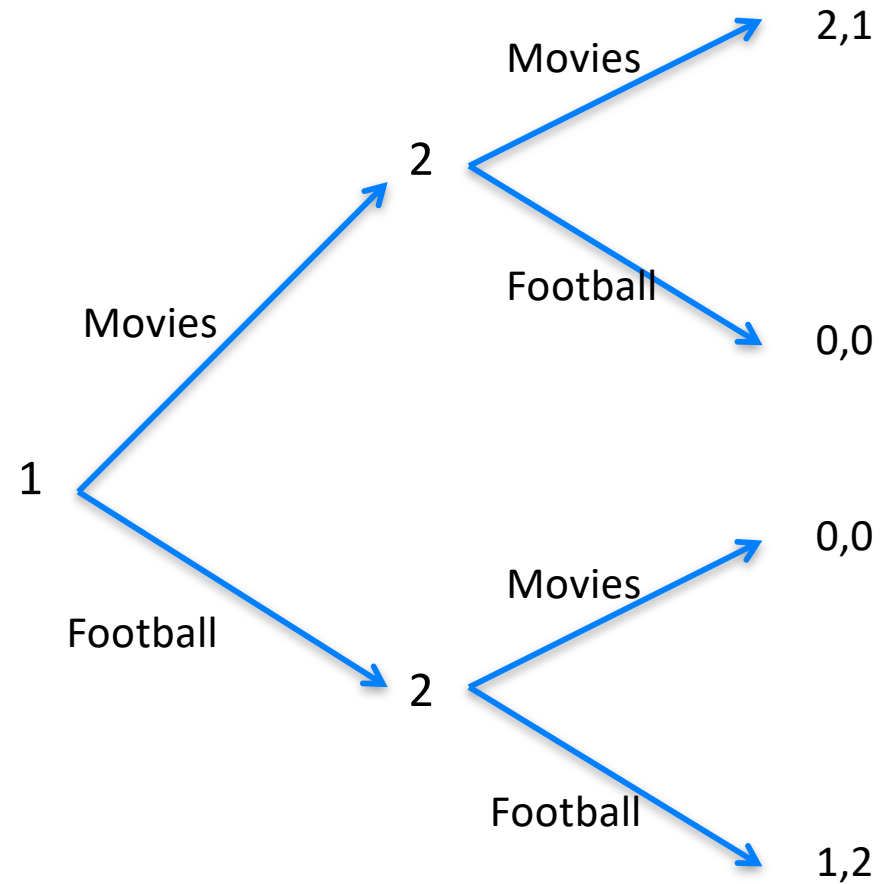
This means that a player that moves first knows the actions played by the others that have led to that point.

Let's review the Battle of the Sexes from a sequential and simultaneous version.

Battle of the Sexes extensive form



SIMULTANEOUS



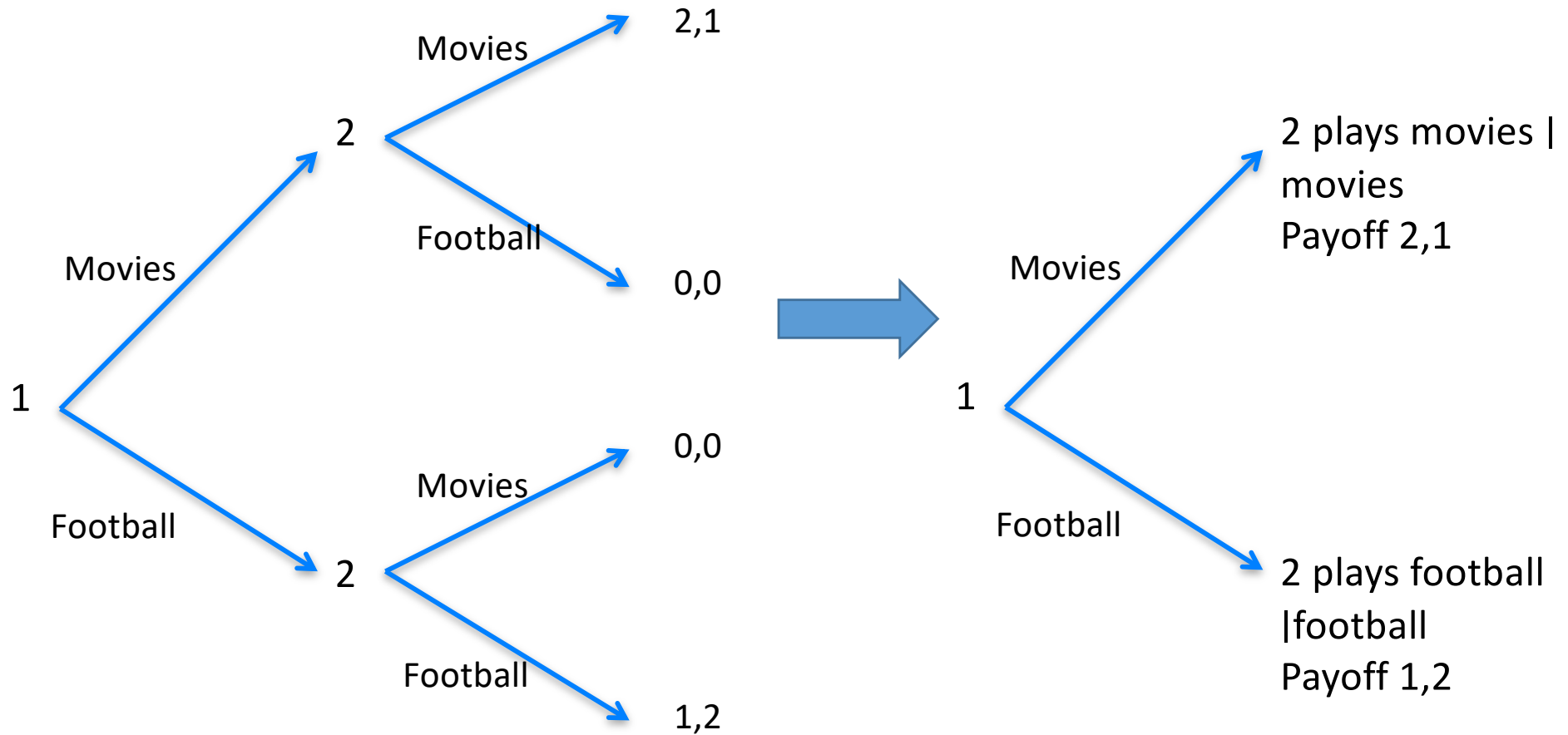
SEQUENTIAL

Player 1: wife, Player 2: husband

Activity 3

Identify the subgames present in the Battle of the Sexes example depicted in the previous slide, for its simultaneous and sequential versions.

Applying Backward Induction



SEQUENTIAL

13. Backward Induction

Backward induction is particularly useful in games that feature many rounds of sequential play. As rounds are added, it quickly becomes too hard to solve for all the Nash equilibria and then to sort through which are subgame – perfect.

With backward induction, an additional round is simply accommodated by adding another iteration of the procedure.

Individual Activity

On a piece of paper, write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it in your own words.

If you have any **questions** about what we discussed today, write them down too. We Will go through them at the beginning of the next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.